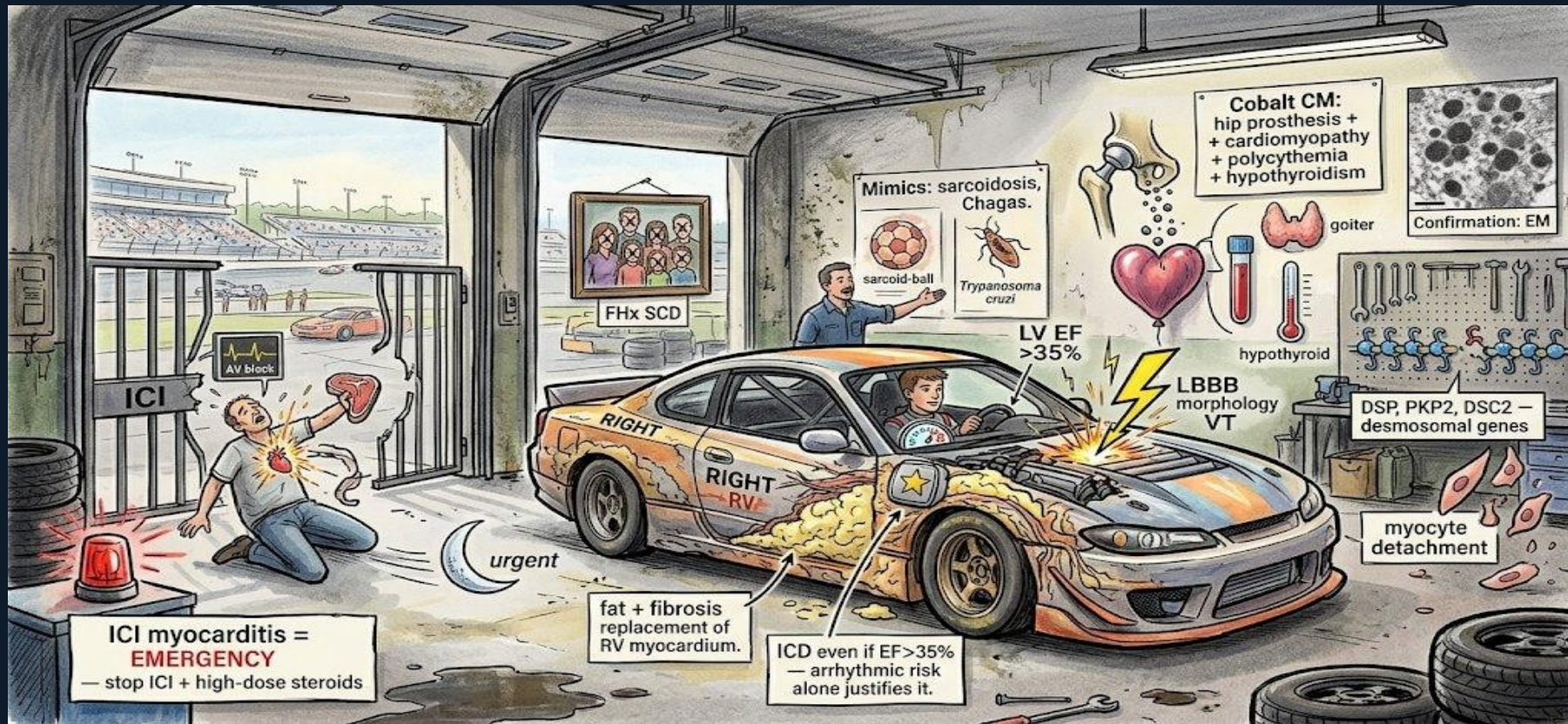
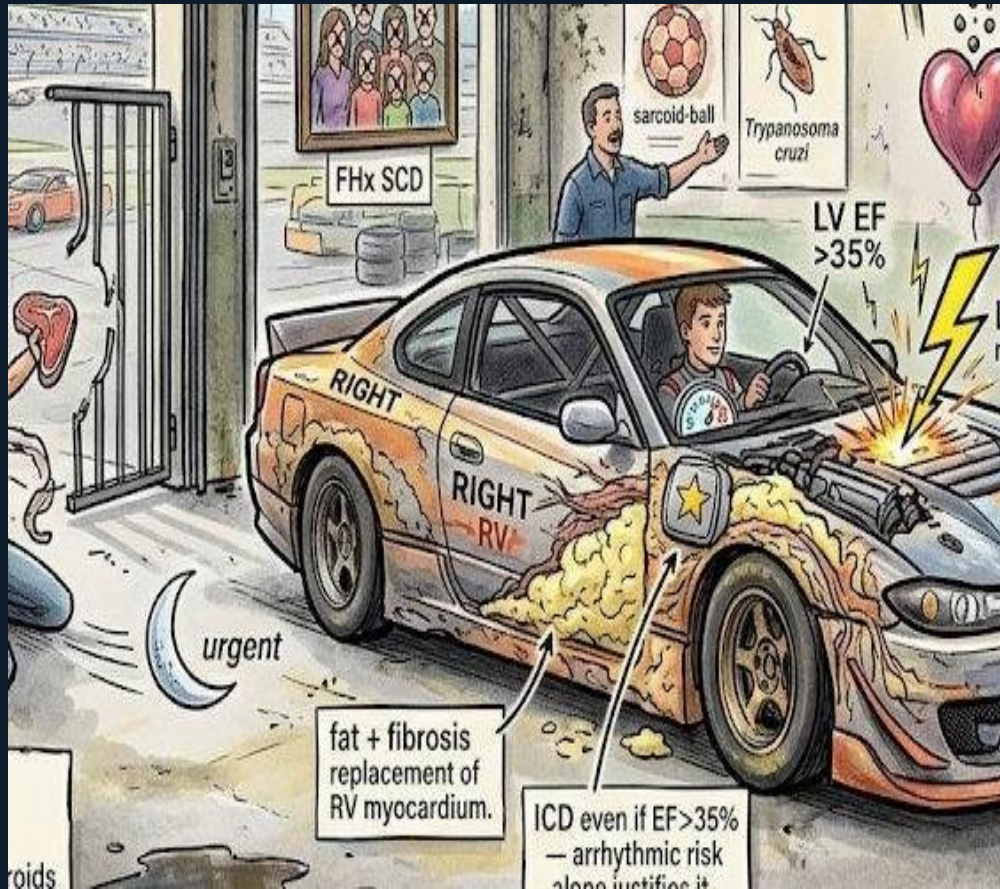


Scene G — ARVC + ICI + Cobalt CMs



Auto Repair Garage • 2 board questions



ARVC: The Crumbling RV Panel

The racing car's right side panel — 'RIGHT RV' — crumbles and is stuffed with fat and fibrous foam. This IS ARVC pathology: desmosomal gene mutations cause myocyte detachment, death, and progressive replacement by adipose tissue and fibrous scar. The resulting scar is an arrhythmogenic substrate. Key board concept: in ARVC, ventricular arrhythmias appear BEFORE EF drops. Arrhythmia is the presenting manifestation, not heart failure. By the time RV dysfunction is apparent, disease is advanced.



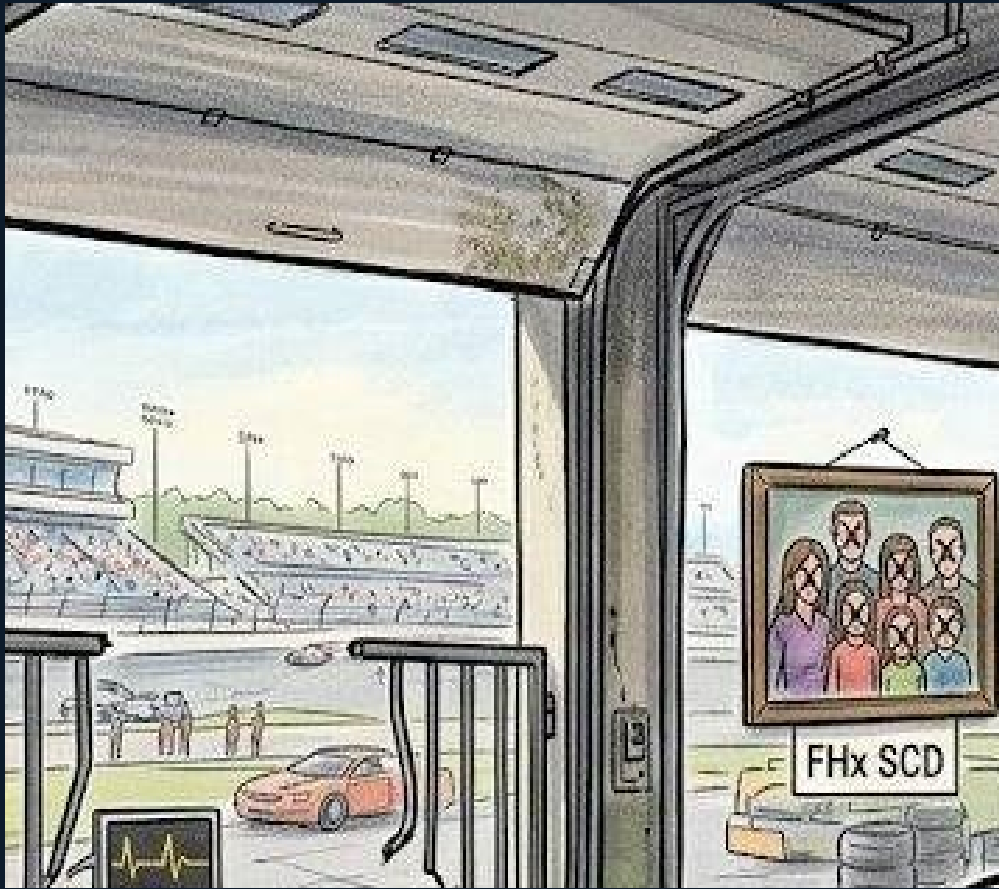
LBBB Morphology VT + ICD >35% EF

VT from the RV produces LBBB morphology on ECG — impulse travels left across the interventricular septum.

ICD gold star on dashboard despite 'LV EF >35%' = critical board rule: in ARVC, ICD may be indicated even with preserved EF when arrhythmic risk is high.

Different from DCM where the threshold is EF $\leq$ 35%.

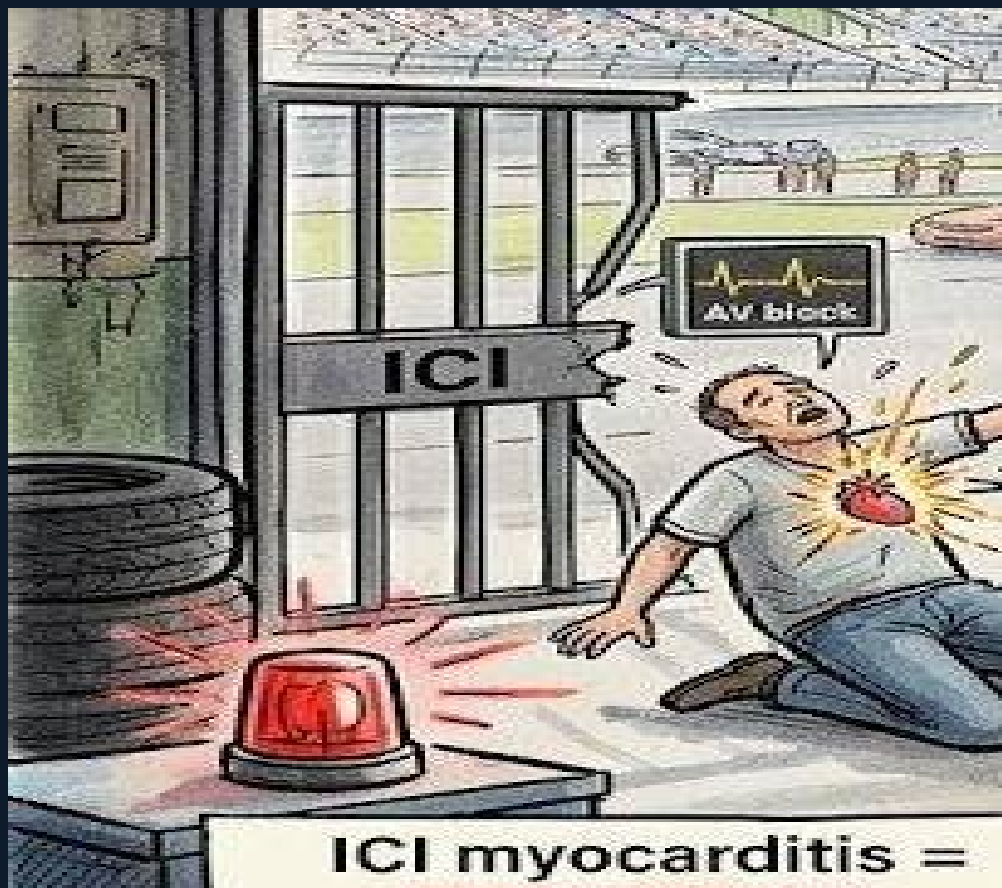
In ARVC, arrhythmic risk (sustained VT, prior arrest, high-risk features) drives the ICD decision, not EF alone.



Scene G — ARVC + ICI + Cobalt | Auto Repair Garage

Family History + Mimics

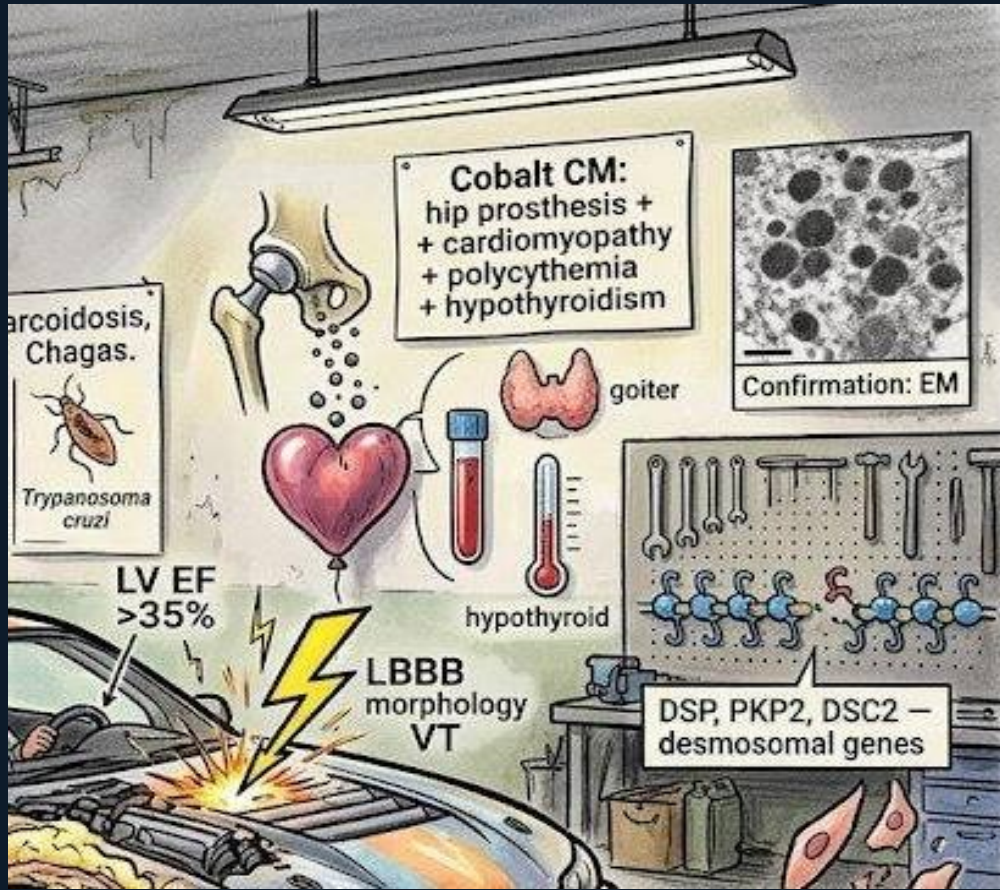
Family portrait with X marks = FHx SCD is a major risk factor. Desmosomal variants follow autosomal dominant inheritance with incomplete penetrance. The mimics poster: cardiac sarcoidosis (sarcoid ball) and Chagas disease (*Trypanosoma cruzi*) can produce identical RV dysfunction and arrhythmias. Both are treatable with different approaches. When ARVC is suspected, sarcoidosis and Chagas must be excluded — CMR and appropriate testing required.



Scene G — ARVC + ICI + Cobalt | Auto Repair Garage

ICI Myocarditis = Emergency

The broken ICI checkpoint gate with collapsing patient encodes checkpoint inhibitor myocarditis. Patients on anti-PD-1, anti-PD-L1, or anti-CTLA-4 therapy can develop acute — often fulminant — myocarditis. Presentation: troponin elevation + new arrhythmia or AV conduction block. Board rule: ICI myocarditis = EMERGENCY — stop immunotherapy immediately + high-dose corticosteroids. Mortality is substantial. Early recognition is critical. The steroid crescent moon rushing in labeled 'urgent' encodes the treatment priority.



Cobalt Cardiomyopathy

Hip prosthesis dripping cobalt into a floppy heart
balloon = cobalt cardiomyopathy. Metal-on-metal hip prostheses release cobalt ions causing:
cardiomyopathy (dilated or restrictive), polycythemia (cobalt has erythropoietin-like effect), hypothyroidism, and goiter. The board vignette: patient with hip prosthesis + new cardiomyopathy + polycythemia + thyroid abnormalities = cobalt toxicity. Confirmation: dense intramitochondrial particles on electron microscopy.



Scene G — ARVC + ICI + Cobalt | Auto Repair Garage

Desmosomal Genes

The interlocking molecular hooks on the tool wall = desmosomes — cell-to-cell adhesion junctions anchoring cardiac myocytes. When desmosomal genes are mutated, myocytes detach under mechanical stress, die, and are replaced by fat and fibrous tissue. Key ARVC genes: DSP (desmoplakin), PKP2 (plakophilin-2), DSC2 (desmocollin-2). PKP2 is most commonly identified. DSP mutations cause a left-dominant ACM variant where the LV is primarily affected, not the RV — an increasingly recognized but underdiagnosed phenotype.

Scene G — Quick Reference

ARVC = fat + fibrosis replaces RV

Desmosomal mutation → myocyte detachment

ICD even if EF >35%

Arrhythmic risk drives decision in ARVC — not EF

Mimics: sarcoidosis, Chagas

Exclude before diagnosing ARVC

Cobalt CM: hip prosthesis + polycythemia

+ hypothyroidism + goiter = cobalt vignette

VT of LBBB morphology = RV origin

Impulse travels left → LBBB pattern on ECG

FHx SCD = major risk factor

Autosomal dominant, incomplete penetrance

ICI myocarditis = EMERGENCY

Stop ICI + high-dose steroids — high mortality

DSP, PKP2, DSC2 — desmosomal genes

PKP2 most common; DSP → LV-dominant variant